

Data Warehouse Distribué avec Apache Hive

Partie 1 – Mise en place de l'environnement (Docker)

```
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker-compose up -d
[+] up 5/5
✓ Network hive          Created
✓ Container postgres    Started
✓ Container metastore   Started
✓ Container hiveserver2 Started
✓ Container superset     Started
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> |
```

```
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker exec -u root -it hiveserver2 bash
root@e97a8b94daa5:/opt/hive# |
```

```
root@e97a8b94daa5:/opt/hive# beeline -u jdbc:hive2://localhost:10000
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/hive/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Connecting to jdbc:hive2://localhost:10000
Connected to: Apache Hive (version 4.0.0)
Driver: Hive JDBC (version 4.0.0)
Transaction isolation: TRANSACTION_REPEATABLE_READ
Beeline version 4.0.0 by Apache Hive
0: jdbc:hive2://localhost:10000> |
```

Partie 2 – Conception d'un entrepôt de données (schéma en étoile)

```
0: jdbc:hive2://localhost:10000> CREATE TABLE dim_client ( id_client INT, nom STRING, ville STRING, pays STRING );
INFO : Compiling command(queryId=hive_20260329152907_cae4de37-9d07-401b-8b50-7605b9b7a92b): CREATE TABLE dim_client ( id_client INT, nom STRING, ville STRI
NG, pays STRING )
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=hive_20260329152907_cae4de37-9d07-401b-8b50-7605b9b7a92b); Time taken: 1.436 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20260329152907_cae4de37-9d07-401b-8b50-7605b9b7a92b): CREATE TABLE dim_client ( id_client INT, nom STRING, ville STRI
NG, pays STRING )
INFO : Starting task [Stage-0:DDL] in serial mode
ERROR : Failed
org.apache.hadoop.hive.ql.metadata.HiveException: AlreadyExistsException(message=Table hive.default.dim_client already exists)
    at org.apache.hadoop.hive.ql.metadata.Hive.createTable(Hive.java:1380) ~[hive-exec-4.0.0.jar:4.0.0]
    at org.apache.hadoop.hive.ql.metadata.Hive.createTable(Hive.java:1388) ~[hive-exec-4.0.0.jar:4.0.0]
    at org.apache.hadoop.hive.ql.ddl.table.create.CreateTableOperation.createTableNonReplaceMode(CreateTableOperation.java:158) ~[hive-exec-4.0.0.jar:4.
0.0]
```

```

0: jdbc:hive2://localhost:10000> CREATE TABLE dim_produit ( id_produit INT, nom_produit STRING, categorie STRING );
INFO : Compiling command(queryId=hive_20260329153212_1f71ff8e-a5ce-45d3-a73d-16ec6aa5b3dc): CREATE TABLE dim_produit ( id_produit INT, nom_produit STRING, categorie STRING )
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=hive_20260329153212_1f71ff8e-a5ce-45d3-a73d-16ec6aa5b3dc); Time taken: 0.01 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20260329153212_1f71ff8e-a5ce-45d3-a73d-16ec6aa5b3dc): CREATE TABLE dim_produit ( id_produit INT, nom_produit STRING, categorie STRING )
INFO : Starting task [Stage-0:DDL] in serial mode
ERROR : Failed

```

```

ERROR : FAILED: Execution Error, return code 40000 from org.apache.hadoop.hive.ql.ddl.DDLTask. AlreadyExistsException(message:Table hive.default.dim_temps already exists)
INFO : Completed executing command(queryId=hive_20260329155027_c4e9c51b-0dc8-4796-b0d7-f228d5d038ef); Time taken: 0.034 seconds
Error: Error while compiling statement: FAILED: Execution Error, return code 40000 from org.apache.hadoop.hive.ql.ddl.DDLTask. AlreadyExistsException(message:Table hive.default.dim_temps already exists) (state=08S01,code=40000)
0: jdbc:hive2://localhost:10000>

```

```

ERROR : FAILED: Execution Error, return code 40000 from org.apache.hadoop.hive.ql.ddl.DDLTask. AlreadyExistsException(message:Table hive.default.ventes already exists)
INFO : Completed executing command(queryId=hive_20260329155152_dbfd3f57-3e6b-4d2c-9f13-0c5a2aa91bed); Time taken: 0.034 seconds
Error: Error while compiling statement: FAILED: Execution Error, return code 40000 from org.apache.hadoop.hive.ql.ddl.DDLTask. AlreadyExistsException(message:Table hive.default.ventes already exists) (state=08S01,code=40000)
0: jdbc:hive2://localhost:10000>

```

Partie 3 – Chargement des données dans Hive

```

0: jdbc:hive2://localhost:10000> SELECT * FROM ventes;
INFO : Compiling command(queryId=hive_20260329155950_63ea31f6-dafb-4f83-a0ad-7ca8aa525261): SELECT * FROM ventes
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:ventes.id_vente, type:int, comment:null), FieldSchema(name:ventes.id_client, type:int, comment:null), FieldSchema(name:ventes.id_produit, type:int, comment:null), FieldSchema(name:ventes.id_temps, type:int, comment:null), FieldSchema(name:ventes.montant, type:float, comment:null)], properties:null)
INFO : Completed compiling command(queryId=hive_20260329155950_63ea31f6-dafb-4f83-a0ad-7ca8aa525261); Time taken: 2.903 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20260329155950_63ea31f6-dafb-4f83-a0ad-7ca8aa525261): SELECT * FROM ventes
INFO : Completed executing command(queryId=hive_20260329155950_63ea31f6-dafb-4f83-a0ad-7ca8aa525261); Time taken: 0.0 seconds

```

ventes.id_vente	ventes.id_client	ventes.id_produit	ventes.id_temps	ventes.montant
1	1	101	202301	150.5
2	2	102	202301	200.0
3	3	103	202301	120.75
4	1	104	202302	180.25
5	4	101	202303	220.3
6	5	102	202303	210.9
7	2	103	202304	185.6
8	3	104	202304	160.8
9	1	101	202305	145.3
10	4	102	202305	230.0

10 rows selected (4.495 seconds)

```
0: jdbc:hive2://localhost:10000>
```

```

0: jdbc:hive2://localhost:10000> SELECT * FROM dim_client
. . . . .> ;
INFO : Compiling command(queryId=hive_20260329160137_48343343-14f3-42a2-b9b2-2cd9a774dabe): SELECT * FROM dim_client
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:dim_client.id_client, type:int, comment:null), FieldSchema(name:dim_client.nom, type:string, comment:null), FieldSchema(name:dim_client.ville, type:string, comment:null), FieldSchema(name:dim_client.pays, type:string, comment:null)], properties:null)
INFO : Completed compiling command(queryId=hive_20260329160137_48343343-14f3-42a2-b9b2-2cd9a774dabe); Time taken: 0.193 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20260329160137_48343343-14f3-42a2-b9b2-2cd9a774dabe): SELECT * FROM dim_client
INFO : Completed executing command(queryId=hive_20260329160137_48343343-14f3-42a2-b9b2-2cd9a774dabe); Time taken: 0.001 seconds

```

dim_client.id_client	dim_client.nom	dim_client.ville	dim_client.pays
1	Ali	Casablanca	Maroc
2	Sanae	Tanger	Maroc
3	Youssef	Fès	Maroc
4	Fatima	Rabat	Maroc
5	Mohamed	Agadir	Maroc

5 rows selected (0.264 seconds)

```
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT * FROM dim_produit;
INFO : Compiling command(queryId=hive_20260329160350_cd670d36-040b-4f8e-a8b0-c54be303a8a8): SELECT * FROM dim_produit
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:dim_produit.id_produit, type:int, comment:null), FieldSchema(name:dim_produit.nom_produit, type:string, comment:null), FieldSchema(name:dim_produit.categorie, type:string, comment:null)], properties:null)
INFO : Completed compiling command(queryId=hive_20260329160350_cd670d36-040b-4f8e-a8b0-c54be303a8a8); Time taken: 0.188 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20260329160350_cd670d36-040b-4f8e-a8b0-c54be303a8a8): SELECT * FROM dim_produit
INFO : Completed executing command(queryId=hive_20260329160350_cd670d36-040b-4f8e-a8b0-c54be303a8a8); Time taken: 0.001 seconds
```

dim_produit.id_produit	dim_produit.nom_produit	dim_produit.categorie
101	Laptop Lenovo	Informatique
102	Smartphone Samsung	Téléphonie
103	Clavier Logitech	Accessoires
104	Écran Dell	Informatique

```
4 rows selected (0.227 seconds)
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT * FROM dim_temps;
INFO : Compiling command(queryId=hive_20260329160451_5089f6d7-48e7-46f0-a289-1305e98cf2a9): SELECT * FROM dim_temps
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:dim_temps.id_temps, type:int, comment:null), FieldSchema(name:dim_temps.jour, type:int, comment:null), FieldSchema(name:dim_temps.mois, type:string, comment:null), FieldSchema(name:dim_temps.annee, type:int, comment:null)], properties:null)
INFO : Completed compiling command(queryId=hive_20260329160451_5089f6d7-48e7-46f0-a289-1305e98cf2a9); Time taken: 0.111 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20260329160451_5089f6d7-48e7-46f0-a289-1305e98cf2a9): SELECT * FROM dim_temps
INFO : Completed executing command(queryId=hive_20260329160451_5089f6d7-48e7-46f0-a289-1305e98cf2a9); Time taken: 0.0 seconds
```

dim_temps.id_temps	dim_temps.jour	dim_temps.mois	dim_temps.annee
202301	1	Janvier	2023
202302	5	Février	2023
202303	12	Mars	2023
202304	20	Avril	2023
202305	28	Mai	2023

```
5 rows selected (0.13 seconds)
0: jdbc:hive2://localhost:10000>
```

Partie 4 – Requêtes OLAP

```
0: jdbc:hive2://localhost:10000> SELECT t.annee, SUM(v.montant) FROM ventes v JOIN dim_temps t ON v.id_temps = t.id_temps GROUP BY t.annee;
INFO : Compiling command(queryId=hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece): SELECT t.annee, SUM(v.montant) FROM ventes v JOIN dim_temps t ON v.id_temps = t.id_temps GROUP BY t.annee
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:t.annee, type:int, comment:null), FieldSchema(name:c1, type:double, comment:null)], properties:null)
INFO : Completed compiling command(queryId=hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece); Time taken: 1.073 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece): SELECT t.annee, SUM(v.montant) FROM ventes v JOIN dim_temps t ON v.id_temps = t.id_temps GROUP BY t.annee
INFO : Query ID = hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece
INFO : Total jobs = 1
INFO : Launching Job 1 out of 1
INFO : Starting task [Stage-1:MAPRED] in serial mode
INFO : Subscribed to counters: [] for queryId: hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece
INFO : Tez session hasn't been created yet. Opening session
INFO : Dag name: SELECT t.annee, SUM(v..... GROUP BY t.annee (Stage-1)
INFO : Setting tez.task.scale.memory.reserve-fraction to 0.30000001192092896
INFO : HS2 Host: [c046d9297ab0], Query ID: [hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece], Dag ID: [dag_1774800394520_0001_1], DAG Session ID: [application_1774800394520_0001]
INFO : Status: Running (Executing on YARN cluster with App id application_1774800394520_0001)
```

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 3	container	SUCCEEDED	1	1	0	0	0	0
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0

```
VERTICES: 03/03 [=====]>>] 100% ELAPSED TIME: 2.08 s
INFO : Completed executing command(queryId=hive_20260329160631_d83367a3-84b2-486b-a46f-f147f1da2ece); Time taken: 5.682 seconds
```

t.annee	c1
2023	1804.4000091552734

```
1 row selected (6.79 seconds)
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT p.nom_produit, SUM(v.montant) AS total FROM ventes v JOIN dim_produit p ON v.id_produit = p.id_produit GROUP BY p.nom_produit ORDER BY total DESC LIMIT 3;
INFO : Compiling command(queryId=hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393): SELECT p.nom_produit, SUM(v.montant) AS total FROM ventes v JOIN dim_produit p ON v.id_produit = p.id_produit GROUP BY p.nom_produit ORDER BY total DESC LIMIT 3
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:p.nom_produit, type:string, comment:null), FieldSchema(name:total, type:double, comment:null)], properties:null)
INFO : Completed compiling command(queryId=hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393); Time taken: 0.356 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393): SELECT p.nom_produit, SUM(v.montant) AS total FROM ventes v JOIN dim_produit p ON v.id_produit = p.id_produit GROUP BY p.nom_produit ORDER BY total DESC LIMIT 3
INFO : Query ID = hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393
INFO : Total jobs = 1
INFO : Launching Job 1 out of 1
INFO : Starting task [Stage-1:MAPRED] in serial mode
INFO : Subscribed to counters: [] for queryId: hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393
INFO : Session is already open
INFO : Dag name: SELECT p.nom_produit,.....otal DESC LIMIT 3 (Stage-1)
INFO : Setting tez.task.scale.memory.reserve-fraction to 0.30000001192092896
INFO : HS2 Host: [c046d9297ab0], Query ID: [hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393], Dag ID: [dag_1774800394520_0001_2], DAG Session ID: [application_1774800394520_0001]
INFO : Status: Running (Executing on YARN cluster with App id application_1774800394520_0001)
```

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 4	container	SUCCEEDED	1	1	0	0	0	0	0
Map 1	container	SUCCEEDED	1	1	0	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0	0
Reducer 3	container	SUCCEEDED	1	1	0	0	0	0	0

VERTICES: 04/04 [=====>>] 100% ELAPSED TIME: 1.23 s

```
INFO : Completed executing command(queryId=hive_20260329160807_f126e2c0-0e90-4df8-b324-7f41cd3b7393); Time taken: 1.325 seconds
```

p.nom_produit	total
Smartphone Samsung	640.8999938964844
Laptop Lenovo	516.1000061035156
Écran Dell	341.0500030517578

3 rows selected (1.698 seconds)

```
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT c.pays, COUNT(*) FROM ventes v JOIN dim_client c ON v.id_client = c.id_client GROUP BY c.pays;
INFO : Compiling command(queryId=hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54): SELECT c.pays, COUNT(*) FROM ventes v JOIN dim_client c ON v.id_client = c.id_client GROUP BY c.pays
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:c.pays, type:string, comment:null), FieldSchema(name:_c1, type:bigint, comment:null)], properties:null)
INFO : Completed compiling command(queryId=hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54); Time taken: 0.379 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54): SELECT c.pays, COUNT(*) FROM ventes v JOIN dim_client c ON v.id_client = c.id_client GROUP BY c.pays
INFO : Query ID = hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54
INFO : Total jobs = 1
INFO : Launching Job 1 out of 1
INFO : Starting task [Stage-1:MAPRED] in serial mode
INFO : Subscribed to counters: [] for queryId: hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54
INFO : Session is already open
INFO : Dag name: SELECT c.pays, COUNT(.....t GROUP BY c.pays (Stage-1)
INFO : Setting tez.task.scale.memory.reserve-fraction to 0.30000001192092896
INFO : Tez session was closed. Reopening...
INFO : Session re-established.
INFO : Session re-established.
INFO : HS2 Host: [c046d9297ab0], Query ID: [hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54], Dag ID: [dag_1774801154118_0001_1], DAG Session ID: [application_1774801154118_0001]
INFO : Status: Running (Executing on YARN cluster with App id application_1774801154118_0001)
```

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 3	container	SUCCEEDED	1	1	0	0	0	0	0
Map 1	container	SUCCEEDED	1	1	0	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0	0

VERTICES: 03/03 [=====>>] 100% ELAPSED TIME: 1.13 s

```
INFO : Completed executing command(queryId=hive_20260329161913_6c053105-b541-42b4-b728-cc8e37ff2c54); Time taken: 1.42 seconds
```

c.pays	_c1
Maroc	10

1 row selected (1.844 seconds)

```
0: jdbc:hive2://localhost:10000>
```

Partie 5 : Visualisation des requêtes Hive avec Superset – Étapes détaillées

Étape 1 – Ajouter le service Superset dans Docker

```
44 >Run Service
45 hiveserver2:
46   image: apache/hive:${HIVE_VERSION}
47   depends_on:
48     - metastore
49   restart: unless-stopped
50   container_name: hiveserver2
51   environment:
52     HIVE_SERVER2_THRIFT_PORT: 10000
53     SERVICE_OPTS: '-Xmx1G -Dhive.metastore.uris=thrift://metastore:9083'
54     IS_RESUME: 'true'
55     SERVICE_NAME: 'hiveserver2'
56   ports:
57     - '10000:10000'
58     - '10002:10002'
59   volumes:
60     - warehouse:/opt/hive/data/warehouse
61   networks:
62     - hive
63 >Run Service
64 superset:
65   image: apache/superset
66   container_name: superset
67   restart: unless-stopped
68   ports:
69     - "8088:8088"
70   environment:
71     SUPERSET_SECRET_KEY: 'thisISaSECRET_key'
72     ADMIN_USERNAME: admin
73     ADMIN_PASSWORD: admin
74     ADMIN_FIRST_NAME: Admin
75     ADMIN_LAST_NAME: User
76     ADMIN_EMAIL: admin@example.com
77   volumes:
78     - superset_home:/app/superset_home
79   depends_on:
80     - hiveserver2
81   command: >
82     /bin/sh -c "
83       superset db upgrade &&
84       superset fab create-admin --username admin --firstname Admin --lastname User --email admin@example.com --password admin &&
85       superset init &&
86       superset run -h 0.0.0.0 -p 8088
87     "
88   networks:
89     - hive
90 volumes:
91   hive-db:
92   warehouse:
93   superset_home:
94 networks:
95   hive:
96     name: hive
97
```


Étape 2 – Installer les outils nécessaires dans le conteneur Superset

1. Ouvrir un terminal root dans le conteneur Superset :

```
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker exec -u root -it superset bash
root@e0bbbd98719b:/app#
```

2. Installer les outils de compilation nécessaires pour les paquets Python natifs :

```
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker exec -u root -it superset bash
root@e0bbbd98719b:/app# apt-get update && apt-get install -y build-essential g++
Get:1 http://deb.debian.org/debian bookworm InRelease [151 kB]
Get:2 http://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:3 http://deb.debian.org/debian-security bookworm-security InRelease [48.0 kB]
Get:4 http://deb.debian.org/debian bookworm/main amd64 Packages [8792 kB]
Get:5 http://deb.debian.org/debian bookworm-updates/main amd64 Packages [6924 B]
Get:6 http://deb.debian.org/debian-security bookworm-security/main amd64 Packages [294 kB]
Fetched 9347 kB in 6s (1660 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu bzip2 cpp cpp-12 dirmngr dpkg-dev fakeroot g++-12 gcc gcc-12 gnupg gnupg-l10n gnupg-utils gpg gpg-agent
  gpg-wks-client gpg-wks-server gpgconf gpgsm libalgorithms-diff-perl libalgorithms-diff-xs-perl libalgorithms-merge-perl libasan8 libassuan0 libatomic1
  libbinutils libc61-0 libc6t0-nobfd0 libc6t0 libdpkg-perl libfakeroot libfile-fcntllock-perl libgcc-12-dev libgdbm-compat4 libgomp1 libgprofng0 libisl23
  libitm1 libjansson4 libksba8 liblocale-gettext-perl liblsan0 libmpc3 libmpfr6 libnpt0 libperl5.36 libquadmath0 libstdc++-12-dev libtsan2 libubsan1 make
  patch perl perl-modules-5.36 pinentry-curses xz-utils
Suggested packages:
  binutils-doc bzip2-doc cpp-doc gcc-12-locales cpp-12-doc dbus-user-session libpam-systemd pinentry-gnome3 tor debian-keyring g++-multilib g++-12-multilib
  gcc-12-doc gcc-multilib manpages-dev autoconf automake libtool flex bison gdb gcc-doc gcc-12-multilib parcimonie xloadimage scd daemon sensible-utils git bzr
  libstdc++-12-doc make-doc ed diffutils-doc perl-doc libterm-readline-gnu-perl | libterm-readline-perl-perl libtap-harness-archive-perl pinentry-doc
The following NEW packages will be installed:
```

```
Setting up gcc (4:12.2.0-3) ...
Setting up g++ (4:12.2.0-3) ...
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode
Setting up build-essential (12.9) ...
Processing triggers for libc-bin (2.36-9+deb12u13) ...
root@e0bbbd98719b:/app#
```

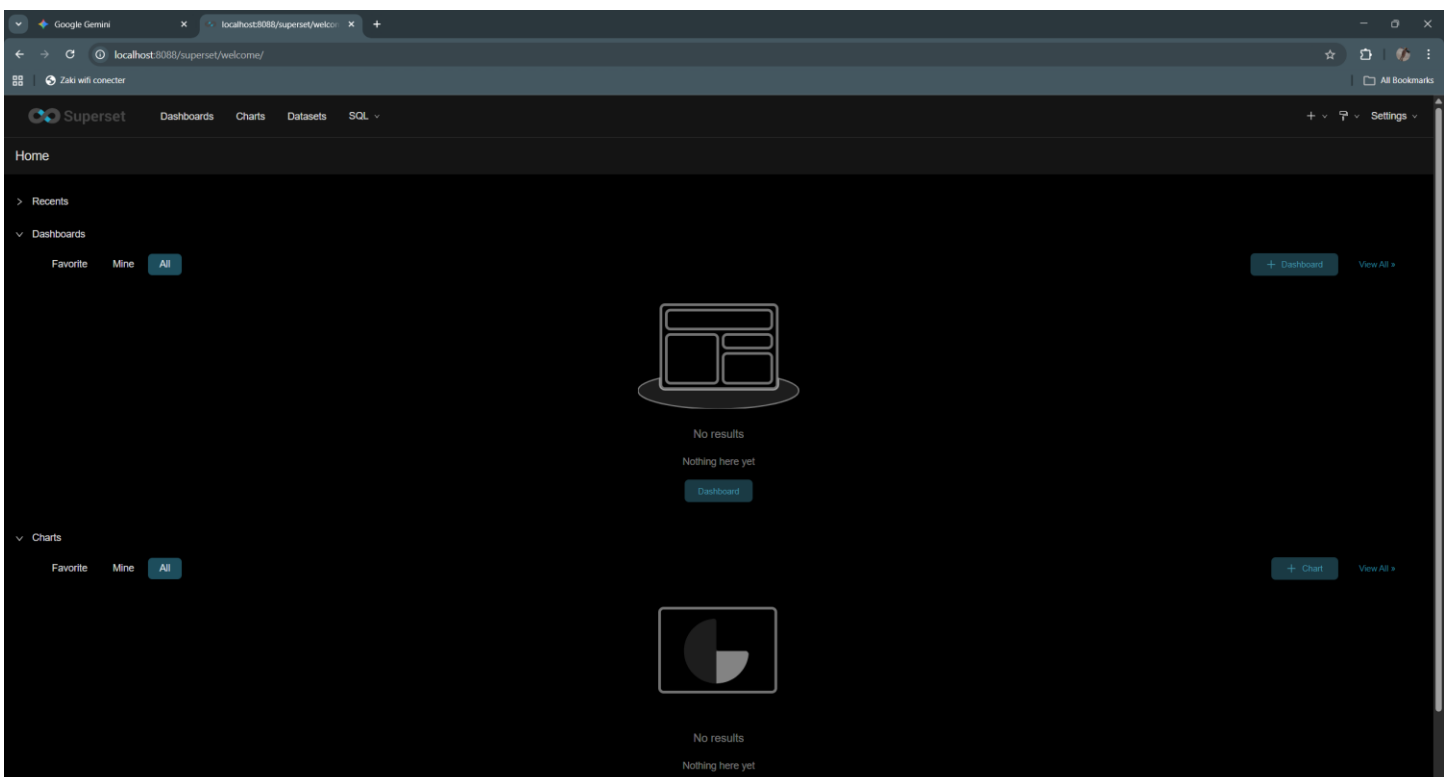
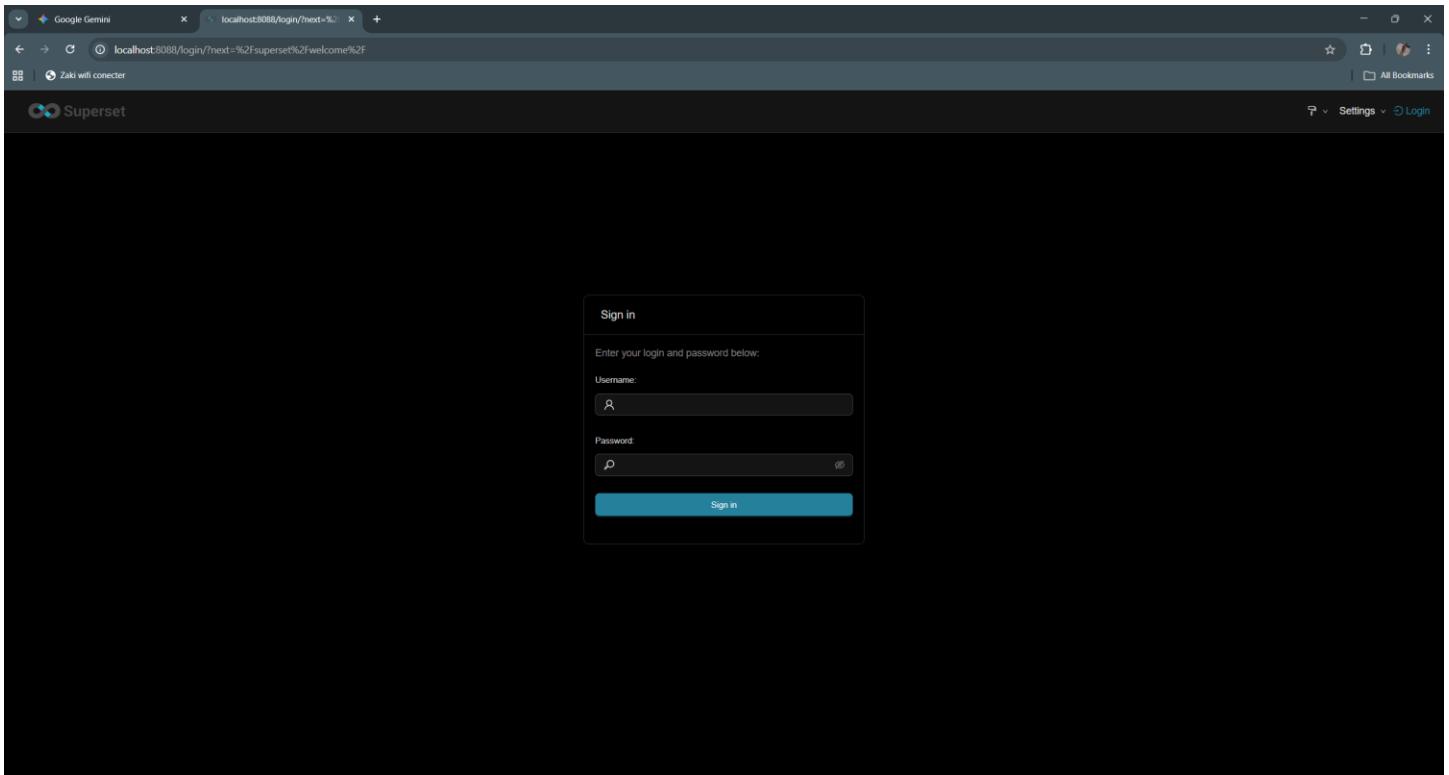
3. Installer les bibliothèques Python permettant à Superset de se connecter à Hive :

```
root@e0bbbd98719b:/app# pip install "pyhive[hive]" thrift sasl thrift-sasl
WARNING: The directory '/app/superset_home/.cache/pip' or its parent directory is not owned or is not writable by the current user. The cache has been disabled.
Check the permissions and owner of that directory. If executing pip with sudo, you should use sudo's -H flag.
Collecting pyhive[hive]
  Downloading PyHive-0.7.0.tar.gz (46 kB)
    46.5/46.5 kB 532.5 kB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Collecting thrift
  Downloading thrift-0.22.0.tar.gz (62 kB)
    62.3/62.3 kB 1.6 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Collecting sasl
  Downloading sasl-0.3.1.tar.gz (44 kB)
    44.7/44.7 kB 2.7 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Collecting thrift-sasl
  Downloading thrift_sasl-0.4.3-py3-none-any.whl (8.3 kB)
Collecting future
```

```
Successfully built thrift sasl pure-sasl pyhive
Installing collected packages: thrift, pure-sasl, six, future, thrift-sasl, sasl, python-dateutil, pyhive
Successfully installed future-1.0.0 pure-sasl-0.6.2 pyhive-0.7.0 python-dateutil-2.9.0.post0 sasl-0.3.1 six-1.17.0 thrift-0.22.0 thrift-sasl-0.4.3
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv

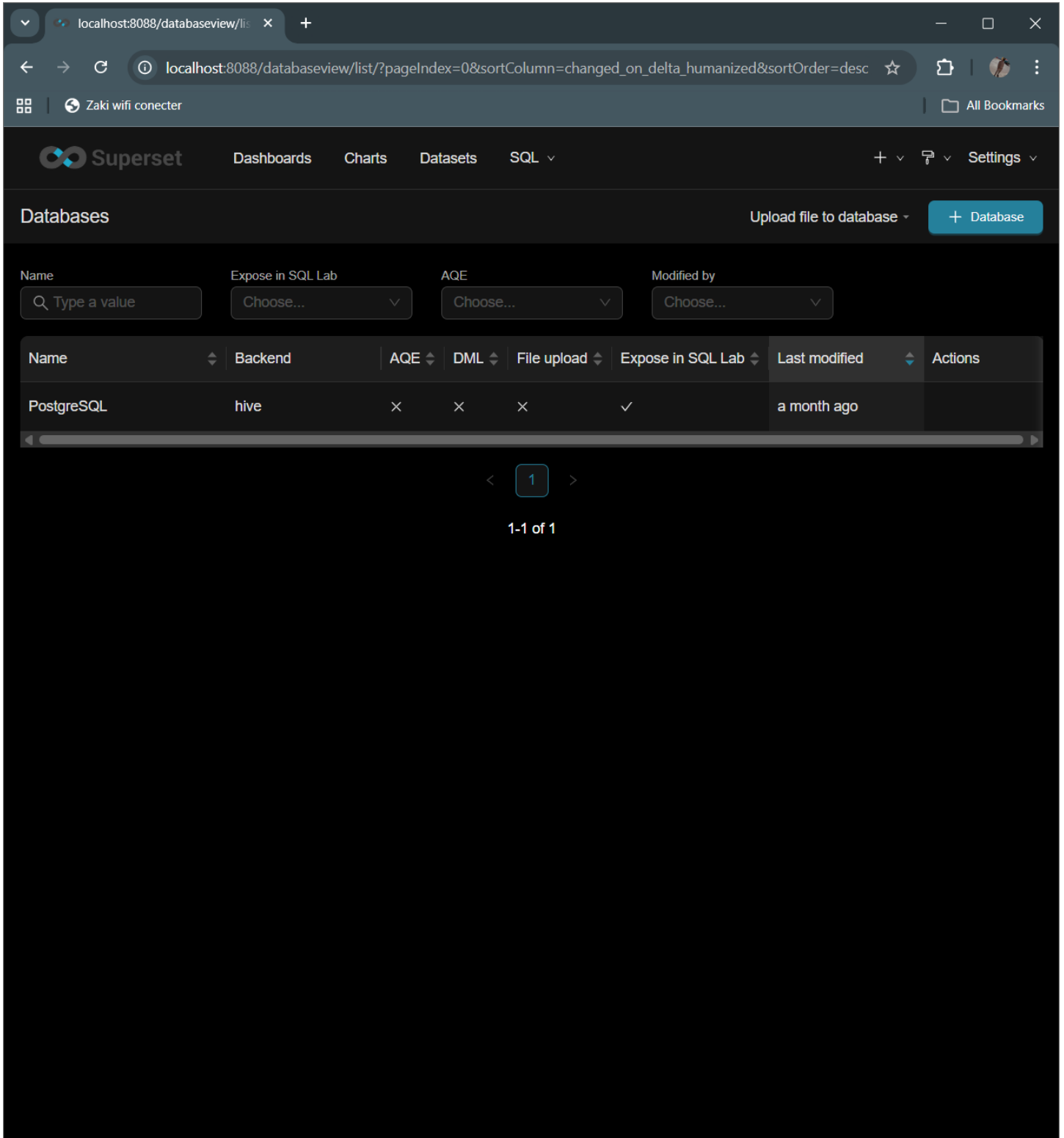
[notice] A new release of pip is available: 23.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
root@e0bbbd98719b:/app#
```

Étape 3 – Démarrer Superset et accéder à l'interface



Étape 4 – Connecter Superset à Hive

1. Dans Superset, aller dans le menu Data > Databases



The screenshot shows the Apache Superset web interface in a browser window. The address bar shows the URL `localhost:8088/databaseview/list/?pageIndex=0&sortColumn=changed_on_delta_humanized&sortOrder=desc`. The Superset logo and navigation menu (Dashboards, Charts, Datasets, SQL) are at the top. The 'Databases' section is active, displaying a table of existing databases. A '+ Database' button is in the top right of the section. The table has columns for Name, Backend, AQE, DML, File upload, Expose in SQL Lab, Last modified, and Actions. One database, 'PostgreSQL', is listed with 'hive' as the backend. Below the table, a pagination control shows '1' of 1 items.

Superset

Dashboards Charts Datasets SQL

Databases Upload file to database + Database

Name Expose in SQL Lab AQE Modified by

Q Type a value Choose... Choose... Choose...

Name	Backend	AQE	DML	File upload	Expose in SQL Lab	Last modified	Actions
PostgreSQL	hive	×	×	×	✓	a month ago	

< 1 >

1-1 of 1

2. Cliquer sur + Database

The screenshot shows the Apache Superset web interface at `localhost:8088/databaseview/list`. A modal window titled "Connect a database" is open, showing "STEP 1 OF 3" and the instruction "Select a database to connect".

Below the instruction, there are two large buttons with database icons: "PostgreSQL" and "SQLite".

Below these buttons, the text "Or choose from a list of other databases we support:" is displayed. Underneath, there is a section titled "Supported databases" with a dropdown menu labeled "Choose a database...".

At the bottom of the modal, there is an information icon and the text "Want to add a new database?". Below this, it states: "Any databases that allow connections via SQL Alchemy URIs can be added. Learn about how to connect a database driver [here](#)." At the very bottom of the modal, there is a link that says "Import database from file".

The background interface shows the "Databases" section with a search bar and a table listing databases. The table has columns for "Name", "Expose", "Backend", "Last modified", and "Actions". One database, "PostgreSQL", is listed with "hive" as the backend and "month ago" as the last modified time.

3. Choisir "Connect this database using a SQLAlchemy URI string"

The screenshot shows the Superset web interface with a modal dialog titled "Connect a database". The dialog is at "STEP 2 OF 2" and "Enter Primary Credentials". It has two tabs: "Basic" (selected) and "Advanced".

In the "Basic" tab, there are two main input fields:

- Display Name ***: A text input containing the value "ventes". Below it is a hint: "Pick a name to help you identify this database."
- SQLAlchemy URI ***: A text input containing the value "hive://hive@hiveserver2:10000/default". Below it is a hint: "Refer to the [SQLAlchemy docs](#) for more information on how to structure your URI."

Below the input fields is a "Test connection" button. Underneath that, a small icon and text say "Connect this database using the dynamic form instead".

A blue information box contains the following text:

Additional fields may be required
Select databases require additional fields to be completed in the Advanced tab to successfully connect the database. Learn what requirements your databases has [here](#).

At the bottom of the dialog are two buttons: "Back" and "Connect". The "Connect" button is currently disabled (greyed out).

4. Dans le champ `SQLAlchemy URI`, entrer :

localhost:8088/databaseview/li: X

localhost:8088/databaseview/list/?pageIndex=0&sortColumn=changed_on_delta_humanized&sortOrder=desc

Zaki wifi conector

All Bookmarks

Superset

DashboardsChartsDatasetsSQL

+ Settings

Databases

Connect a database

STEP 2 OF 2

Enter Primary Credentials

Need help? Learn how to connect your database [here](#).

BasicAdvanced

Display Name *

ventas

Pick a name to help you identify this database.

SQLAlchemy URI *

hive://hive@hiveserver2:10000/default

Refer to the [SQLAlchemy docs](#) for more information on how to structure your URI.

Test connection

Connect this database using the dynamic form instead

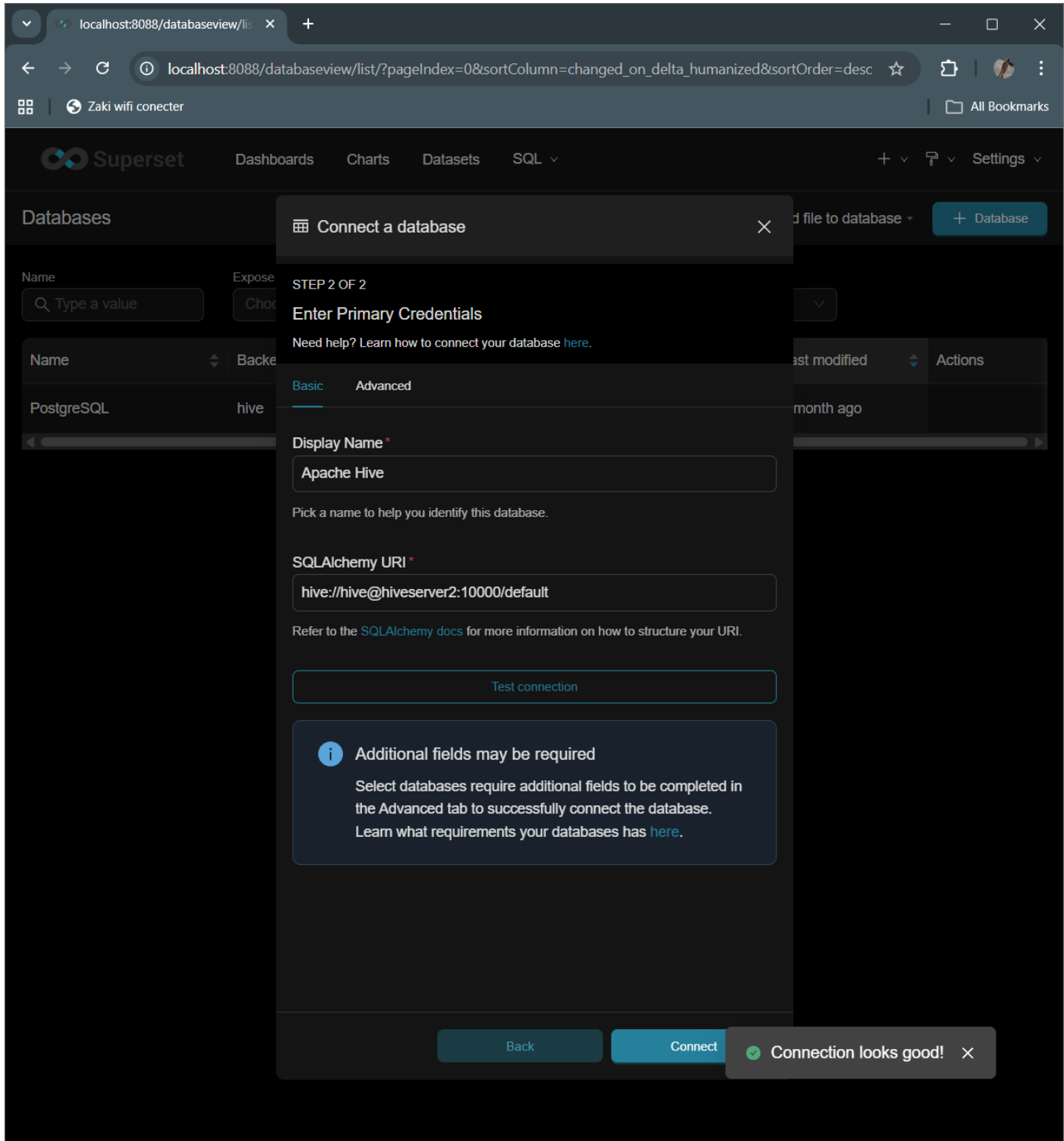
Additional fields may be required

Select databases require additional fields to be completed in the Advanced tab to successfully connect the database. Learn what requirements your databases has [here](#).

BackConnect

5. Cliquer sur Test Connection

6. Si tout est correct, le message *It looks good!* s'affiche



7. Cliquer sur Save

localhost:8088/databaseview/

localhost:8088/databaseview/list/?pageIndex=0&sortColumn=changed_on_delta_humanized&sortOrder=desc

Zaki wifi connecter

All Bookmarks

Superset

DashboardsChartsDatasetsSQL

+ Settings

Databases

Upload file to database + Database

Name

Expose in SQL Lab

AQE

Modified by

Q Type a value

Choose...

Choose...

Choose...

Name	Backend	AQE	DML	File upload	Expose in SQL Lab	Last modified	Actions
Apache Hive	hive	x	x	x	✓	now	
PostgreSQL	hive	x	x	x	✓	a month ago	

< 1 >

1-2 of 2

Étape 5 – Créer une requête Hive

1. Aller dans SQL Lab > SQL Editor

The screenshot shows the Apache Superset web interface in a browser. The URL bar indicates the connection is to `localhost:8088/sqlab/`. The Superset logo and navigation menu (Dashboards, Charts, Datasets, SQL) are at the top. The 'SQL' tab is active, showing a query editor with the following SQL code:

```
1 SELECT t.annee, SUM(v.montant) AS total
2 FROM ventes v
3 JOIN dim_temps t ON v.id_temps = t.id_temps
4 GROUP BY t.annee
```

On the left sidebar, the 'Database' dropdown is set to 'hive PostgreSQL', and the 'Schema' dropdown is set to 'default'. Below these, there is a search bar for table schemas. At the bottom of the editor, there are buttons for 'Run', 'LIMIT: 1 000', a timer showing '00:00:00.00', 'Save', and 'Copy link'. Below the editor, there are tabs for 'Results' and 'Query history'. The 'Results' tab is currently empty, displaying a message: 'Run a query to display results' and 'There is currently no information to display.'

2. Sélectionner la base de données Hive et le schéma default

The screenshot displays the Apache Superset SQL Lab interface in a web browser. The browser's address bar shows the URL `localhost:8088/sqlab/`. The Superset navigation bar at the top includes links for Dashboards, Charts, Datasets, and SQL. The main interface is divided into three primary sections:

- Left Panel:** Contains configuration options for the query. The 'Database' dropdown is set to 'hive' (Apache Hive). The 'Schema' dropdown is set to 'default'. Below these, there is a 'See table schema' section with a search input field.
- SQL Editor:** A text area containing the following SQL query:

```
1 SELECT t.annee, SUM(v.montant) AS total
2 FROM ventes v
3 JOIN dim_temps t ON v.id_temps = t.id_temps
4 GROUP BY t.annee
```
- Right Panel:** Features a 'Run' button, a 'LIMIT' dropdown set to '1 000', a timer showing '00:00:00.00', and buttons for 'Save' and 'Copy link'. Below the editor, there are tabs for 'Results' and 'Query history'. The 'Results' tab is active, showing a message: 'Run a query to display results' and 'There is currently no information to display.'

3. Exemple de requête OLAP :

The screenshot displays the Apache Superset SQL Lab interface in a web browser. The browser's address bar shows the URL `localhost:8088/sqlab/`. The Superset header includes navigation links for Dashboards, Charts, Datasets, and SQL, along with user settings and a Zaki wifi connecter extension.

The interface is divided into three main sections:

- Left Panel:** Contains configuration options for the query. The **Database** is set to `hive` (Apache Hive). The **Schema** is set to `default`. A section for **See table schema** includes a search bar with the placeholder text "Select table or type to search tables".
- SQL Editor:** Displays the following SQL query:

```
1 SELECT t.annee, SUM(v.montant) AS total
2 FROM ventes v
3 JOIN dim_temps t ON v.id_temps = t.id_temps
4 GROUP BY t.annee
```
- Right Panel:** Features a **Run** button, a **LIMIT** dropdown set to `1 000`, a timer showing `00:00:00.00`, and buttons for **Save** and **Copy link**. Below these are tabs for **Results** and **Query history**. The **Results** tab is active, showing a message: "Run a query to display results" and "There is currently no information to display."

4. Cliquer sur Run pour exécuter la requête

The screenshot shows the Apache Superset SQL Lab interface in a web browser. The browser address bar shows `localhost:8088/sqlab/`. The Superset header includes navigation links for Dashboards, Charts, Datasets, and SQL. The main interface is divided into three sections:

- Left Panel:** Contains configuration for the database and schema. The 'Database' dropdown is set to 'hive' (Apache Hive). The 'Schema' dropdown is set to 'default'. Below these is a search bar for 'See table schema'.
- Center Panel:** Displays the SQL query being executed:

```
1 SELECT t.annee, SUM(v.montant) AS total
2 FROM ventes v
3 JOIN dim_temps t ON v.id_temps = t.id_temps
4 GROUP BY t.annee
```
- Right Panel:** Shows the execution controls and results. The 'Run' button is highlighted. Below it, the 'Results' tab is active, showing a table with one row of data.

Execution controls include a 'LIMIT' of 1 000, a timer showing 00:00:16.276, and buttons for 'Save' and 'Copy link'.

Below the query, there are buttons for 'Create chart', 'Download to CSV', 'Copy to Clipboard', and 'Filter results'. The query text is repeated in a smaller font, and the number of rows is indicated as '1 row'.

t.annee	total
2023	1804.4000091552734

5. Cliquer sur Explore pour visualiser les résultat

Superset

Dashboards Charts Datasets SQL

Add the name of the chart

Save

Chart Source

SELECT t.annee AS annee, SUM...

Search Metrics & Columns

Create a dataset to edit or add columns and metrics.

Metrics

Showing 0 of 0 items

Columns

Showing 2 of 2 items

annee

total

Data

Customize

Table 4k

View all charts

Query

Query mode

Aggregate Raw records

Columns

annee

total

+ Drop columns here or click

Filters

+ Drop columns/metrics here or click

Ordering

Select ...

Server pagination

Row limit

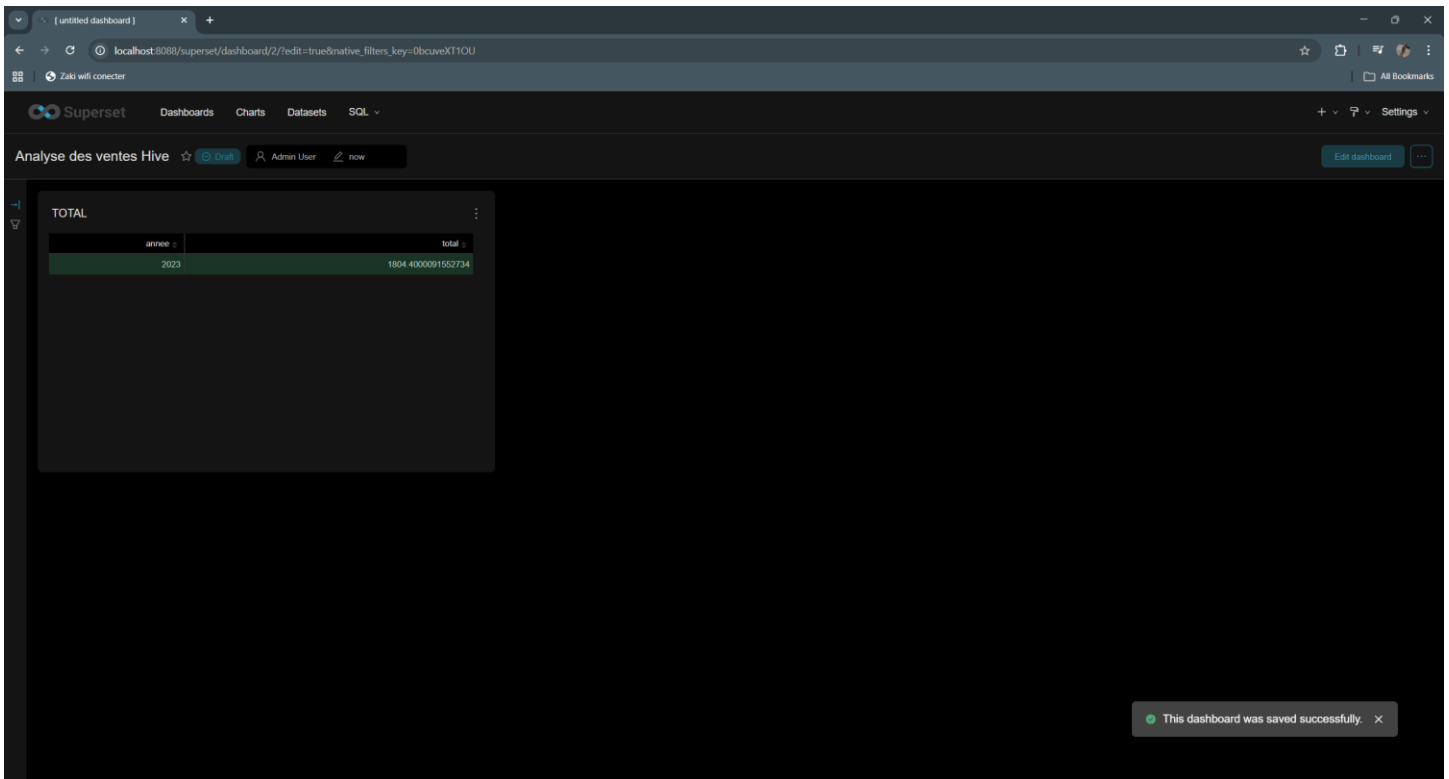
1000

Update chart

Results Samples

annee	
2023	1804

Étape 6 – Créer un tableau de bord



Partie 6 – Optimisations avancées

```
0: jdbc:hive2://localhost:10000> SET hive.enforce.bucketing = true;
No rows affected (0.04 seconds)
0: jdbc:hive2://localhost:10000> SET hive.enforce.sorting = true;
No rows affected (0.007 seconds)
0: jdbc:hive2://localhost:10000>
```

Partie 6 – Travaux Pratiques

```
0: jdbc:hive2://localhost:10000> -- Création de la tableCREATE TABLE dim_magasin ( id_magasin INT, nom_magasin STRING, ville STRING) STORED AS PARQUET;-- Insertion de données de testINSERT INTO dim_magasin VALUES (10, 'Techno Store Casa', 'Casablanca'),(20, 'Digital Shop Rabat', 'Rabat');
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT c.pays, SUM(v.montant) FROM ventes v JOIN dim_client c ON v.id_client = c.id_client GROUP BY c.pays;
INFO : Compiling command(queryId=hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdae826c5b): SELECT c.pays, SUM(v.montant) FROM ventes v JOIN dim_client c ON v.id_client = c.id_client G
ROUP BY c.pays
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:c.pays, type:string, comment:null), FieldSchema(name:c1, type:double, comment:null)], properties:null)
INFO : Completed compiling command(queryId=hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdae826c5b); Time taken: 0.313 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdae826c5b): SELECT c.pays, SUM(v.montant) FROM ventes v JOIN dim_client c ON v.id_client = c.id_client G
ROUP BY c.pays
INFO : Query ID = hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdae826c5b
INFO : Total jobs = 1
INFO : Launching Job 1 out of 1
INFO : Starting task [Stage-1:MAPRED] in serial mode
INFO : Subscribed to counters: [] for queryId: hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdae826c5b
INFO : Session is already open
INFO : Dag name: SELECT c.pays, SUM(v.....t GROUP BY c.pays (Stage-1)
INFO : Setting tez.task.scale.memory.reserve-fraction to 0.30000001102092896
INFO : HS2 Host: [c046d9297ab0], Query ID: [hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdae826c5b], Dag ID: [dag_1774822169877_0001_2], DAG Session ID: [application_1774822169877_00
01]
INFO : Status: Running (Executing on YARN cluster with App id application_1774822169877_0001)

-----
VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 3 ..... container  SUCCEEDED    1         1         0         0         0         0
Map 1 ..... container  SUCCEEDED    1         1         0         0         0         0
Reducer 2 ..... container  SUCCEEDED    1         1         0         0         0         0
-----
VERTICES: 03/03 [=====]>>>] 100% ELAPSED TIME: 1.02 s
-----
INFO : Completed executing command(queryId=hive_20260329221401_5c82b991-cf34-4e15-ac41-cfdae826c5b); Time taken: 1.105 seconds
+-----+
| c.pays |      c1 |
+-----+
| Maroc  | 1804.4000091552734 |
+-----+
1 row selected (1.453 seconds)
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> SELECT pays, total_ca FROM ca_par_pays_et_annee;
INFO : Compiling command(queryId=hive_20260329221449_61105195-7cb5-42cd-a936-c7051a019944): SELECT pays, total_ca FROM ca_par_pays_et_annee
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:pays, type:string, comment:null), FieldSchema(name:total_ca, type:double, comment:null)], properties:null)
INFO : Completed compiling command(queryId=hive_20260329221449_61105195-7cb5-42cd-a936-c7051a019944); Time taken: 0.214 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20260329221449_61105195-7cb5-42cd-a936-c7051a019944): SELECT pays, total_ca FROM ca_par_pays_et_annee
INFO : Completed executing command(queryId=hive_20260329221449_61105195-7cb5-42cd-a936-c7051a019944); Time taken: 0.0 seconds
+-----+
| pays |      total_ca |
+-----+
| Maroc | 1804.4000091552734 |
+-----+
1 row selected (0.336 seconds)
0: jdbc:hive2://localhost:10000>
```

```
0: jdbc:hive2://localhost:10000> DESCRIBE FORMATTED ventes;
INFO : Compiling command(queryId=hive_20260329221759_28a05040-90b5-4739-82ab-4762882d3492): DESCRIBE FORMATTED ventes
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:col_name, type:string, comment:from deserializer), FieldSchema(name:data_type, type:string, comment:from deserializer), FieldSchema(name:comment, type:string, comment:from deserializer)], properties:nu
ll)
INFO : Completed compiling command(queryId=hive_20260329221759_28a05040-90b5-4739-82ab-4762882d3492); Time taken: 0.04 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20260329221759_28a05040-90b5-4739-82ab-4762882d3492): DESCRIBE FORMATTED ventes
INFO : Starting task [Stage-0:DDL] in serial mode
INFO : Completed executing command(queryId=hive_20260329221759_28a05040-90b5-4739-82ab-4762882d3492); Time taken: 0.086 seconds
+-----+
| col_name | data_type | comment |
+-----+
# id_vente | int |
# id_client | int |
# id_produit | int |
# id_temps | int |
# montant | float |
# | NULL | NULL |
# Detailed Table Information | NULL | NULL |
# Database: | default | NULL |
# OwnerType: | USER | NULL |
# Owner: | hive | NULL |
# CreateTime: | Mon Feb 16 17:51:18 UTC 2026 | NULL |
# LastAccessTime: | UNKNOWN | NULL |
# Retention: | 0 | NULL |
# Location: | file:/opt/hive/data/warehouse/ventes | NULL |
# Table type: | EXTERNAL_TABLE | NULL |
# Table Parameters: | NULL | NULL |
# COLUMN_STATS_ACCURATE | {"BASIC_STATS":true,"COLUMN_STATS":{"id_client":true,"id_produit":true,"id_temps":true,"id_vente":true,"montant":true}} |
# EXTERNAL | TRUE |
# TRANSLATED_TO_EXTERNAL | TRUE |
# bucketing_version | 2 |
# external_table.purge | TRUE |
# numFiles | 1 |
# numRows | 10 |
# rawDataSize | 360 |
# totalSize | 1330 |
# transient_lastDdlTime | 1771264437 |
# | NULL | NULL |
# Storage Information | NULL | NULL |
# SerDe Library: | org.apache.hadoop.hive.q1.io.parquet.serde.ParquetHiveSerDe | NULL |
# InputFormat: | org.apache.hadoop.hive.q1.io.parquet.MapredParquetInputFormat | NULL |
# OutputFormat: | org.apache.hadoop.hive.q1.io.parquet.MapredParquetOutputFormat | NULL |
# Compressed: | NO | NULL |
# Num Buckets: | -1 | NULL |
# Bucket Columns: | [] | NULL |
# Sort Columns: | [] | NULL |
# Storage Desc Params: | NULL | NULL |
# serialization.format | 1 |
+-----+
37 rows selected (0.154 seconds)
0: jdbc:hive2://localhost:10000>
```

```
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker exec -it hiveserver2 du -sh /opt/hive/data/warehouse/ventes
12K /opt/hive/data/warehouse/ventes
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker exec -it hiveserver2 du -sh /opt/hive/data/warehouse/ventes_part
4.0K /opt/hive/data/warehouse/ventes_part
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive> docker exec -it hiveserver2 du -sh /opt/hive/data/warehouse/ventes_bucketed
4.0K /opt/hive/data/warehouse/ventes_bucketed
PS C:\Users\Administrator\Desktop\BigData\BigData\Data Warehouse Distribué avec Apache Hive\cluster_hive>
```